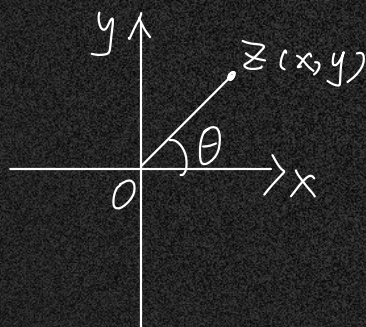


一、复数

$$i^{4n} = 1, \quad i^{4n+1} = i, \quad i^{4n+2} = -1, \quad i^{4n+3} = -i.$$



$$\text{辐角 } \text{Arg } z = \theta \quad \theta \in (-\pi, \pi]$$

$$\text{主辐角 } \arg z$$

($z \neq 0$) $z=0$ 时无意义.

$$\text{Arg } z = \begin{cases} \text{无穷多个} \\ \arg z + 2k\pi \\ \theta + 2k\pi \end{cases} \quad (k=0, \pm 1, \dots)$$

$$\star \star \arg z = \begin{cases} \arctan \frac{y}{x} & (\text{一、四象限}) \\ \arctan \frac{y}{x} + \pi & (\text{二}) \\ \arctan \frac{y}{x} - \pi & (\text{三}) \end{cases}$$

复数的三角表示.

$$[z = x + iy = r \cos \theta + i r \sin \theta = r \cdot e^{i\theta}]$$

欧拉公式

$$|z| = r, \quad e^{i\theta} = \cos \theta + i \sin \theta, \quad \theta = \arg z \text{ 或 } \text{Arg } z$$

三角形式

$$\sin \alpha + i \cos \alpha \Rightarrow \sin \alpha = \cos(\frac{\pi}{2} - \alpha), \quad \cos \alpha = \sin(\frac{\pi}{2} - \alpha) \Rightarrow \cos(\frac{\pi}{2} - \alpha) + i \sin(\frac{\pi}{2} - \alpha)$$

$$|z_1 \cdot z_2| = r_1 r_2 = |z_1| \cdot |z_2| \quad \text{Arg}(z_1 \cdot z_2) = \theta_1 + \theta_2 = \text{Arg } z_1 + \text{Arg } z_2$$

$$|z^n| = |z|^n, \quad \text{Arg}(z^n) = n \text{Arg } z.$$

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$|z_1| + |z_2| \geq |z_1 + z_2|, \quad |z_1| - |z_2| \leq |z_1 - z_2|$$

$$\text{方根: } z^n = |z|^n \cdot e^{i n \theta} \quad \theta = \arg z + 2k\pi \quad (k=0, 1, 2, \dots)$$

$\star \star$ 次方根有 n 个根

$$|z|^{\frac{1}{n}} \cdot e^{i \frac{\theta}{n}} = |z|^{\frac{1}{n}} \cdot e^{i \frac{(\arg z + 2k\pi)}{n}} \quad (k=0, 1, \dots, n-1)$$